

Fungal Infections - Summary

Key Symptoms: depends on affected area

- **Vaginal Candidiasis:** discussed in 'Women's Health' lecture
- **Oral Candidiasis:** Cotton mouth, loss of taste, redness, white patches/plaques on tongue, gums, throat or buccal mucosa, which can be removed with vigorous rubbing but leave red/bleeding sites
- **Esophageal Candidiasis:** Dysphagia, odynophagia, retrosternal chest pain, fever, plaques

Risk factors

- Local: steroids and antibiotics, dentures, anticholinergic AE, local factors (immunocompromised)
- Systemic: drugs (cytotoxics, immunosuppressants, CS, PPIs), neonates/advanced age, diabetes, HIV/AIDS, nutritional deficiencies, radiotherapy/chemotherapy, malignancies

Most common causing-pathogens

- **Yeasts:** Candida spp., Cryptococcus spp.
- **Moulds:** Aspergillus spp., Zygomycetes, Tinea spp. (dermatophytes)
- **Dimorphic:** Histoplasma spp., Blastomyces spp.

Pharmacological Treatment

- **Oral Candidiasis:** Nystatin oral suspension **OR** clotrimazole (troches)
 - **Alternatives:** Itraconazole, voriconazole/posaconazole (refractory), amphotericin B (severe or refractory; IV) **OR** echinocandins (severe or refractory; IV)
- **Esophageal Candidiasis:** Fluconazole (tablet; preferred) or itraconazole (solution)
- **Tinea infections:**
 - **1st line (except for tinea capitis):** topical azoles, terbinafine 1%, tolnaftate 1%, ciclopirox 1%
 - Duration of terbinafine is shorter than that of other agents
 - **Tinea capitis:** systemic therapy is required; oral terbinafine

Additional Key Points

- Esophageal candidiasis must be treated systemically, if a patient has trouble swallowing → IV
 - Other scenarios when systemic therapy is preferred:
 - Pts at high risk of disseminated systemic or invasive candidiasis
 - Pts who cannot tolerate topical agents due to dry mouth or swallowing issues
 - Pts with oral/esophageal candidiasis refractory to topical treatment
 - Severe odynophagia
- **Drugs of choice for mucocutaneous candidiasis:** voriconazole (IV, PO) or posaconazole (PO), fluconazole (IV, PO), itraconazole (PO)
- **Azoles** (e.g., Fluconazole, itraconazole, voriconazole)
 - Newer azoles like voriconazole, posaconazole, isavuconazole overcome the limited efficacy of fluconazole against Aspergillus and other moulds along with non-albicans Candida spp C. krusei and C. glabrata. + improved absorption, tolerability and safety compared to itraconazole
 - Inhibitors of CYP450 3A4 (itra, posa, vori, fluc), 2C9/19 (fluc, vori): see slide 27
- **Echinocandins** (Caspofungin, micafungin, anidulafungin – IV only) – slides 31-34
 - Not effective against zygomycetes, Cryptococcus neoformans and Fusarium species
 - Requires dose adjustment in hepatic impairment but not renal impairment
 - Low propensity for drug interactions
- **Polyenes** (e.g., Nystatin, amphotericin B)
 - Nystatin is less effective than oral azoles; effective in oral candidiasis
 - Amphotericin B requires monitoring of electrolytes and renal function as it is nephrotoxic (do not combine with other nephrotoxic drugs), widest spectrum of activity amongst all antifungals
- LFT required for oral antifungals